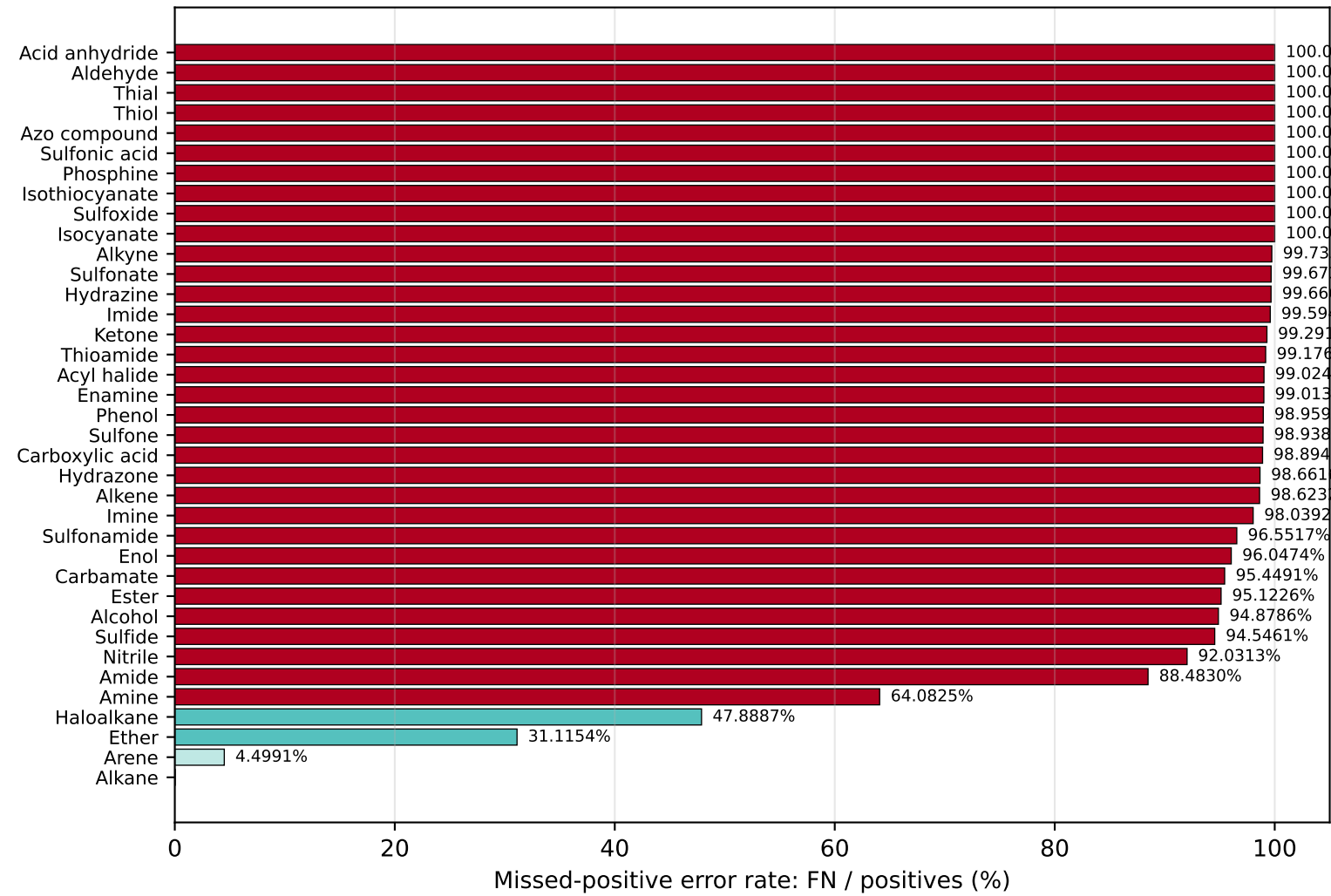
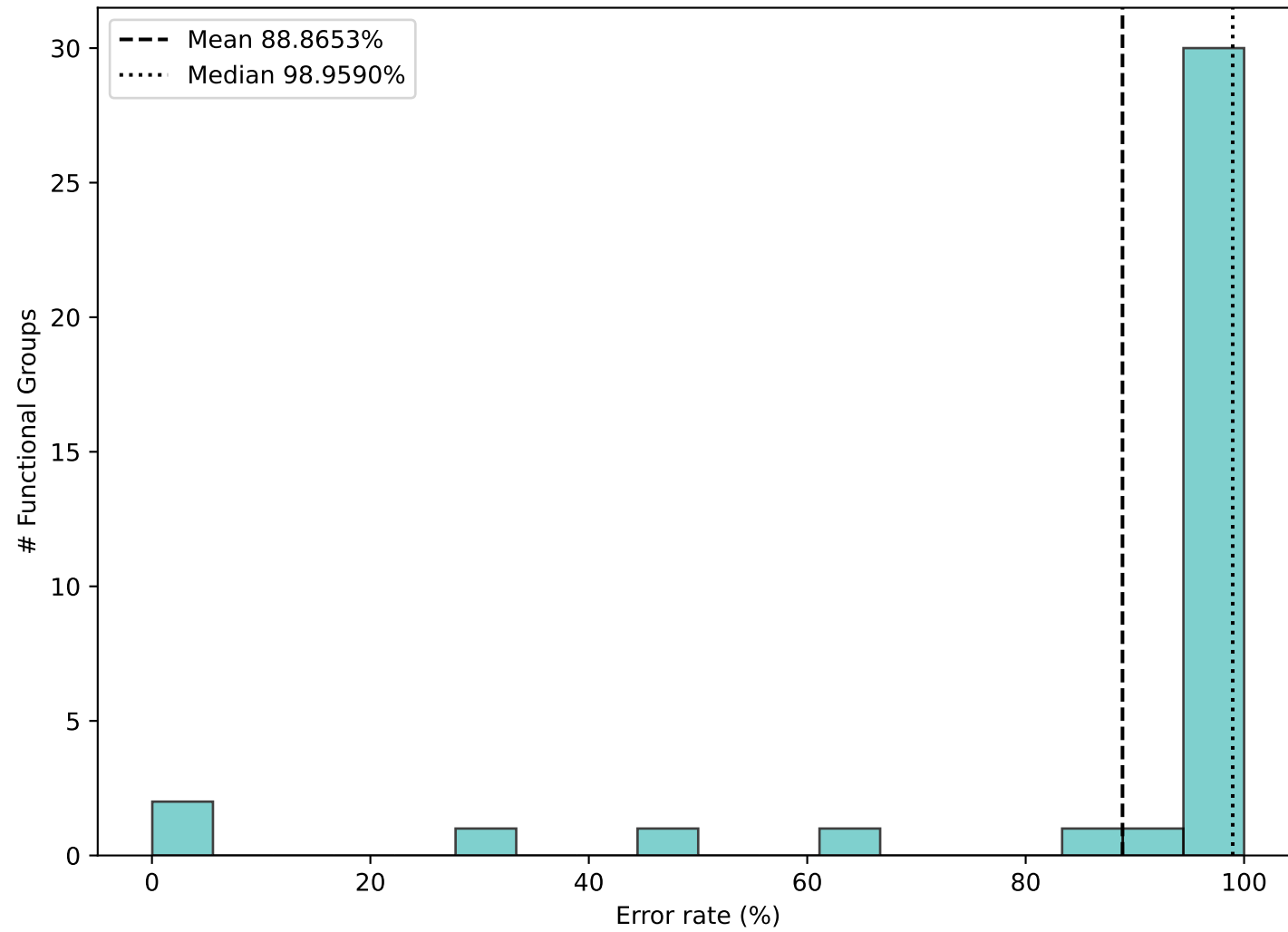


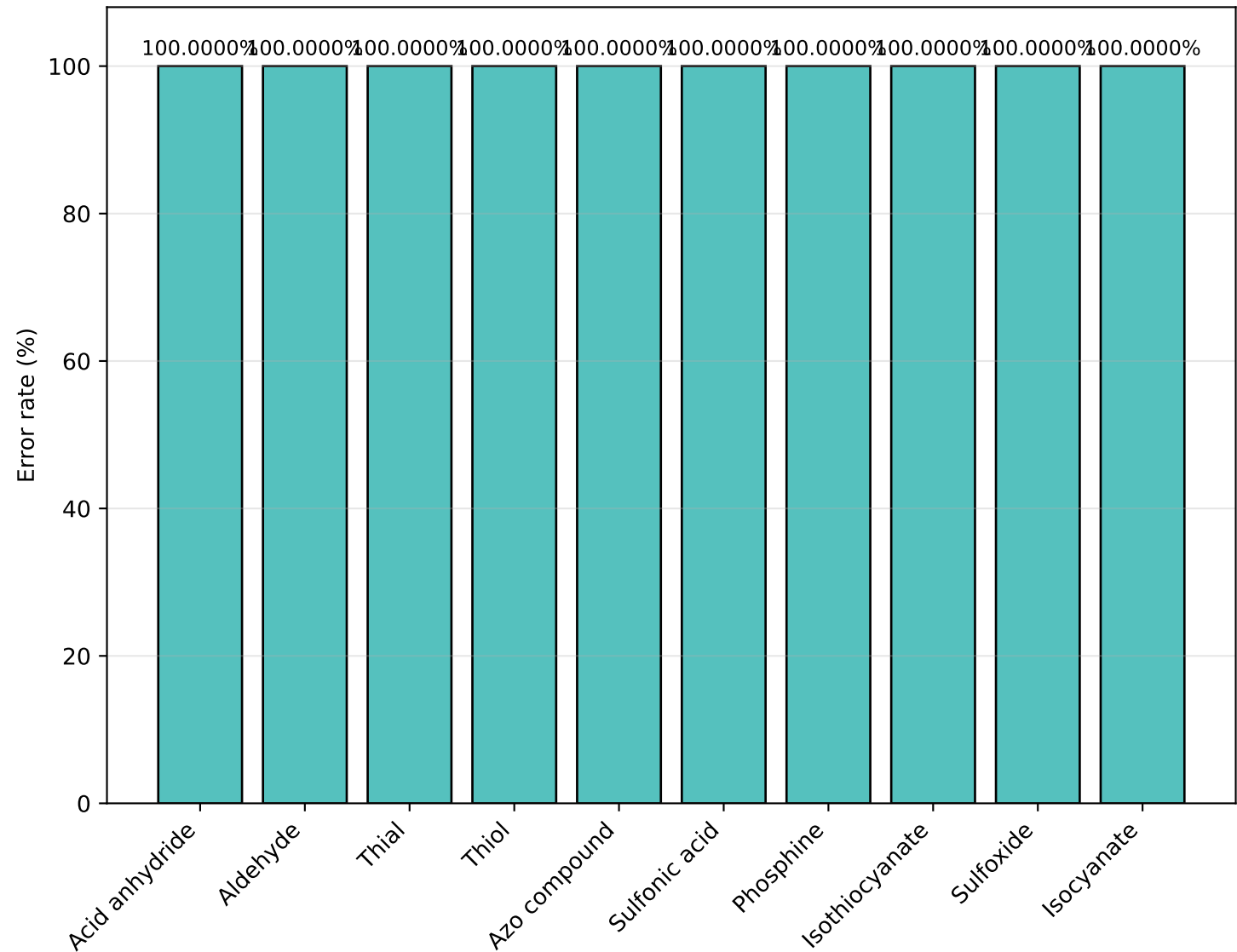
CNN MSMS+ Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



CNN MSMS+ VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.6426
F1 macro: 0.1136
Precision micro: 0.8026
Recall micro: 0.5358
Hamming accuracy: 0.9230

Error-rate statistics:

Mean: 88.8653%
Median: 98.9590%
Std: 25.6126%
Max: 100.0000% (Acid anhydride)
Min: 0.0314% (Alkane)

Problem classes:

Severe (>50%): 33 groups
High (20-50%): 2 groups
Moderate (5-20%): 0 groups
Low (<=5%): 2 groups